

Molar Concentration

It is another quantity which helps in comparing the conductivities of electrolytes. It is defined as : **the conductance of all ions produced by one mole (one gram-molecular weight) of an electrolyte when dissolved in a certain volume V cc.**

Molar conductance is denoted by μ . Its value is obtained by multiplying the specific conductance, κ , by the volume in cc containing one mole of the electrolyte.

Thus,

Molar conductance, $\mu = \kappa \times V$ where V is the volume of the solution in cc containing one mole of the electrolyte.

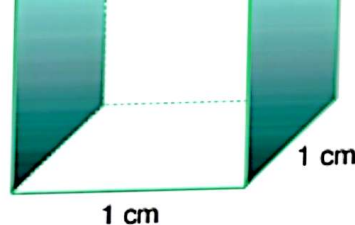
Units of Molar Concentration

Since

$$\begin{aligned}\kappa &= \frac{1}{R} \times \frac{l}{A} \\ \mu &= \frac{1}{R} \times \frac{l}{A} \times V \\ &= \frac{1}{\text{ohm}} \times \frac{\text{cm}}{\text{cm}^2} \times \frac{\text{cm}^3}{\text{mol}}\end{aligned}$$

Specific Conductance

It is evident that a substance which offers very little resistance to the flow of current allows more current to pass through it. Thus the power of a substance to conduct electricity or conductivity is the converse of resistance. The reciprocal of specific resistance is termed **Specific conductance** or **Specific conductivity**.



■ **Figure 24.3**
Diagrammatic illustration of definition of specific conductance.

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It is defined as : **the conductance of one centimetre cube (cc) of a solution of an electrolyte.**

The specific conductance is denoted by the symbol κ (kappa). Thus,

$$\kappa = \frac{1}{\rho} = \frac{1}{R} \times \frac{l}{A}$$

Units of Specific conductance

Specific conductance is generally expressed in reciprocal ohms (r.o) or **mhos** or **ohm⁻¹**. Its unit can be derived as follows :

$$\begin{aligned}\kappa &= \frac{1}{A} \times \frac{l}{R} = \frac{1}{\text{ohm}} \times \frac{\text{cm}}{\text{cm}^2} \\ &= \text{ohm}^{-1}\text{cm}^{-1}\end{aligned}$$

The internationally recommended unit for ohm⁻¹ (or mho) is **Siemens, S**. When S is used, the conductance is expressed as **S cm⁻¹**. It may be noted that Siemens is not a plural, the unit is named after Sir William Siemens—a noted electrical engineer.

The specific conductance increases with : (i) ionic concentration, and (ii) speeds of the ions concerned.

In measuring the specific conductance of the aqueous solution of an electrolyte, the volume of water in which a certain amount of the electrolyte is dissolved is always measured in cubic centimeters (cc) and this is known as **dilution**. If the volume of a solution is V_{cc} , the specific conductance of the solution is written as κ .

Equivalent Conductance

It is defined as **the conductance of an electrolyte obtained by dissolving one gram-equivalent of it in V cc of water.**



SOLVED PROBLEM 2. The resistance of decinormal solution of a salt occupying a volume between two platinum electrodes 1.80 cm apart and 5.4 cm² in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.

SOLUTION

Here $l = 1.80 \text{ cm}$ and $A = 5.4$

$$\therefore \text{cell constant } x = \frac{l}{A} = \frac{1.80}{5.4} = \frac{1}{3}$$

$$\text{observed conductance} = \frac{1}{32} \text{ mhos}$$

Since the solution is *N*/10, $V = 10,000 \text{ ml}$

Now, specific conductance = $x \times \text{obs. conductance}$

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or

$$\kappa = \frac{1}{3} \times \frac{1}{32} = \frac{1}{96} \text{ mhos}$$

$$\Lambda = \kappa \times V$$

$$= \frac{1}{96} \times 10,000$$

$$= 104.1 \text{ mhos cm}^2 \text{ equiv}^{-1}$$

Example 2. A cell of known area being filled with a 0.02 molar solution of KCl at 25°C

FARADAY'S LAWS OF ELECTROLYSIS

Michael Faraday studied the quantitative aspect of electrolysis. He discovered that there exists a definite relationship between the amounts of products liberated at the electrodes and the quantity of electricity used in the process. In 1834, he formulated two laws which are known as **Faraday's Laws of Electrolysis**. These are :

First Law

The amount of a given product liberated at an electrode during electrolysis is directly proportional to the quantity of electricity which passes through the electrolyte solution.

Second Law

When the same quantity of electricity passes through solutions of different electrolytes, the amounts of the substances liberated at the electrodes are directly proportional to their chemical equivalents.

Definition of Electrochemical equivalent in light of First Law

If m is the mass of substance (in grams) deposited on electrode by passing Q coulombs of electricity, then

$$m \propto Q \quad \text{(First Law)}$$

We know that $Q = I \times t$

where I is the strength of current in amperes and t is the time in second for which the current has been passed.

Therefore, $m \propto I \times t$

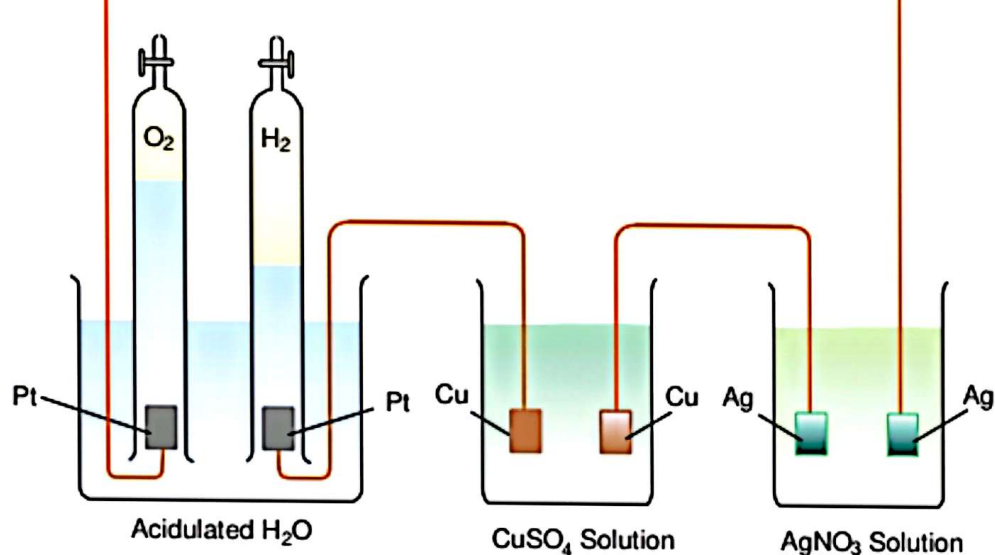
or $m = Z \times I \times t$

where Z is the constant known as the Electrochemical equivalent of the substance (electrolyte).

If $I = 1$ ampere and $t = 1$ second, then

$$m = Z$$

Thus, the electrochemical equivalent is the amount of a substance deposited by 1 ampere current passing for 1 second (*i.e.*, one coulomb).



■ **Figure 24.2**

Illustrating Faraday's Second Law of Electrolysis.

To verify the law, let us take an arrangement of the type shown in Fig. 24.2. Pass the same quantity of electricity through the three coulometers (the term 'coulometer' is now in practice

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replaced by the older term 'voltmeter') containing solution of dilute H_2SO_4 , CuSO_4 and AgNO_3 respectively. These coulometers are fitted with platinum, copper and silver electrodes as shown in Fig. 24.2. The masses of hydrogen, copper and silver liberated/deposited at the respective cathodes are in the ratio of their equivalent weights. That is,

$$\frac{\text{mass of hydrogen liberated}}{\text{mass of copper deposited}} = \frac{\text{Eq. Wt. of hydrogen}}{\text{Eq. Wt. of copper}}$$

and

$$\frac{\text{mass of copper deposited}}{\text{mass of silver deposited}} = \frac{\text{Eq. Wt. of copper}}{\text{Eq. Wt. of silver}}$$

From this experiment, we can calculate the mass of hydrogen, copper and silver liberated at their respective cathodes by one coulomb of electricity. We find these are always :

$$\text{Hydrogen} = 0.00001036 \text{ g}$$

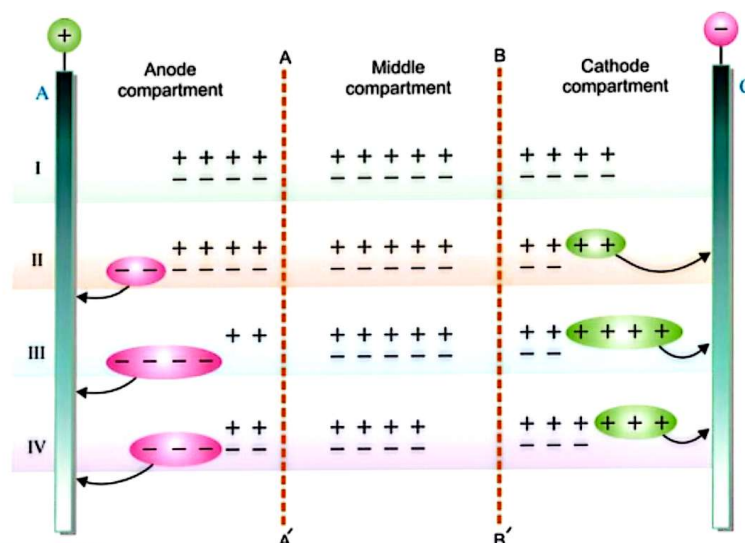
$$\text{Copper} = 31.78 \times 0.00001036 = 0.0003292 \text{ g}$$

$$\text{Silver} = 107.88 \times 0.00001036 = 0.001118 \text{ g}$$

Since the equivalent weights of hydrogen, copper and silver are 1, 31.78 and 107.88 respectively, it follows that **the chemical equivalents are proportional to the chemical equivalents (or equivalent weights).**

anode. Similarly, the speed of an anion moving away from the cathode will be proportional to the fall of concentration of anions around the cathode. Hittorf studied such changes experimentally and gave a general rule known as the **Hittorf's Rule**. It states that : **the loss of concentration around any electrode is proportional to the speed of the ion moving away from it**. Hittorf's Rule may be illustrated by the following scheme.

In Fig. 25.4, A is anode and C is cathode. AA' and BB' are two imaginary planes which divide the cell into three compartments, **the anode compartment, the middle compartment and the cathode compartment**. The sign (+) represents a cation while the sign (–) represents an anion.



■ Figure 25.4

Scheme showing that the loss of concentration around any electrode is proportional to the speed of the ion moving away from it.

Before electrolysis, let there be 13 ion-pairs in the cell. The number of ion-pairs in the two outer compartments is 4 each and there are 5 ion-pairs in the middle compartment (Position I). Now let us consider the following cases:

(i) Let the anions alone be capable of movement

When, say, two anions have moved towards the anode, we get the position as shown in II. The cations have not moved at all. But inspite of that the number of discharged anions and cations is the same viz., 2. *The concentration in the anode compartment, however, has not altered while in the cathode compartment it has fallen by two ion-pairs.*

(ii) Let the anions and cations move at the same rate

When two ions of each type have crossed over towards the opposite electrodes, we get the condition as shown in III. The number of discharged anions and cations is the same viz., 4. *The concentration of both the anode compartment and the cathode compartment has fallen to the same extent viz, by two ion-pairs.*

(iii) Let the cations move at twice the speed of the anions

In this case, when cations have moved to the cathode compartment, one anion passes into the anode compartment. The state of affairs is shown in IV. The total number of discharged anions and cations is again the same viz., 3. Although the concentration in the cathode compartment has fallen by one ion-pair, the concentration in the anode compartment decreases by two ion-pairs.

It is evident from the above considerations that ions are always discharged in equivalent amounts on the opposite electrodes. It is really due to the difference in the speeds of anions and cations that the changes in concentration around electrodes are caused. We also conclude that **the loss in concentration around any electrode is proportional to the speed of the ion moving away from it**.

This is **Hittorf's Rule**. We can write the expression :

$$\frac{\text{Fall around anode}}{\text{Fall around cathode}} = \frac{\text{Speed of cation}}{\text{Speed of anion}} = \frac{v_+}{v_-}$$

where v_+ and v_- is the speed of cations and anions respectively.

WHAT IS TRANSPORT NUMBER ?

During electrolysis the current is carried by the anions and the cations. **The fraction of the total current carried by the cation or the anion is termed its Transport number or Hittorf's number.** If v_+ represents the speed of migration of the cation and v_- that of the anion,

$$\text{the transport number of cation} = \frac{v_+}{v_+ + v_-}$$

$$\text{the transport number of anion} = \frac{v_-}{v_+ + v_-}$$

The transport number of the cation is represented by t and that of the anion by t_- .

Thus
$$t_+ = \frac{v_+}{v_+ + v_-} \quad \text{and} \quad t_- = \frac{v_-}{v_+ + v_-}$$

or
$$\frac{t_+}{t_-} = \frac{v_+}{v_-} \quad \text{and} \quad t_+ + t_- = 1$$

If the speed ratio v_+/v_- be denoted by r , we have,

$$r = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+}$$

and
$$t_- = \frac{1}{1 + r}$$

SOLVED PROBLEM 1. A solution of silver nitrate containing 12.14 g of silver in 50 ml of solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

SOLUTION

Weight of Ag in 50 ml of the solution before electrolysis = 12.14 g

Weight of Ag in 50 ml of the solution after electrolysis = 11.55 g

$\therefore$ Fall in concentration of Ag = 12.14 – 11.55

$$= 0.59 \text{ g}$$

$$= 0.0055 \text{ g eq}$$

Weight of Ag deposited in silver coulometer = 1.25 g

$$= \frac{1.25}{108} \text{ g eq}$$

$$= 0.0116$$

$$\begin{aligned} \text{Hence, transport number of } \text{Ag}^+ (t_{\text{Ag}^+}) &= \frac{\text{Fall in conc. around anode}}{\text{No. of g eqvt deposited in silver coulometer}} \\ &= \frac{0.0055}{0.0116} \\ &= 0.474 \end{aligned}$$

$$\begin{aligned} \text{and Transport number of } \text{NO}_3^- \text{ ion } (t_{\text{NO}_3^-}) &= 1 - 0.474 \\ &= 0.526 \end{aligned}$$

SOLVED PROBLEM 2. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

SOLUTION

Wt. of copper in the anode liquid before electrolysis = 0.79 g

Wt. of copper in the anode liquid after electrolysis = 0.91 g

Increase in weight = 0.91 – 0.79

$$= 0.12 \text{ g}$$

Increase in weight of copper cathode in the coulometer = 0.153 g

This means that if no copper had migrated from the anode, increase in weight would have been 0.153 g.

But actual increase = 0.12

Fall in concentration due to migration of Cu^{2+} ions = 0.153 – 0.12

$$= 0.033$$

$$\begin{aligned} \therefore \text{Transport number of } \text{Cu}^{2+} \text{ ion} &= \frac{0.033}{0.153} \\ &= 0.215 \end{aligned}$$

$$\begin{aligned} \text{and Transport number of } \text{SO}_4^{2-} \text{ ion} &= (1 - 0.215) \\ &= 0.785 \end{aligned}$$

A system with at least one dimension (length, width, or thickness) of the dispersed particles in the range $10 \text{ \AA}$ to $2,000 \text{ \AA}$, is classed as a colloidal dispersion.

TYPES OF COLLOIDAL SYSTEMS

As we have seen above, a colloidal system is made of two phases. The substance distributed as the colloidal particles is called the **Dispersed phase**. The second continuous phase in which the colloidal particles are dispersed is called the **Dispersion medium**. For example, for a colloidal solution of copper in water, copper particles constitute the dispersed phase and water the dispersion medium.

As stated above, a colloidal system is made of a dispersed phase and the dispersion medium. Because either the dispersed phase or the dispersion medium can be a gas, liquid or solid, there are eight types of colloidal systems possible. A colloidal dispersion of one gas in another is not possible since the two gases would give a homogeneous molecular mixture.

The various types of colloidal systems are listed in Table. 22.1.

In this chapter we will restrict our study mainly to the colloidal systems which consist of a solid substance dispersed in a liquid. These are frequently referred to as **Sols** or **Colloidal solution**. The colloidal solutions in water as the dispersion medium are termed **Hydrosols** or **Aquasols**. When the dispersions medium is alcohol or benzene, the sols are referred to as **Alcosols** and **Benzosols** respectively.

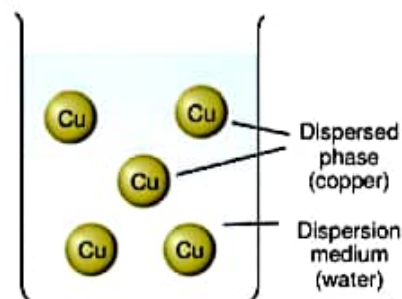


Figure 22.3
A colloidal system of copper in water

TABLE 22.1. TYPES OF COLLOIDAL SYSTEMS

Type Name	Dispersed Phase	Dispersion medium	Examples
Foam	gas	liquid	whipped cream, shaving cream, soda-water
Solid foam	gas	solid	froth cork, pumice stone, foam rubber
Aerosol	liquid	gas	for, mist, clouds
Emulsion	liquid	liquid	milk, hair cream
Solid emulsion (gel)	liquid	solid	butter, cheese
Smoke	solid	gas	dust, soot in air
Sol	solid	liquid	paint, ink, colloidal gold
Solid sol	solid	solid	ruby glass (gold dispersed in glass), alloys.

LYOPHILIC AND LYOPHOBIC SOLS OR COLLOIDS

Sols are colloidal systems in which a solid is dispersed in a liquid.

These can be subdivided into two classes :

- Lyophilic sols (solvent-loving)
- Lyophobic sols (solvent-hating)

Lyophilic sols are those in which the dispersed phase exhibits a definite affinity for the medium or the solvent.

positive charge.

COMPARISON OF LYOPHILIC AND LYOPHOBIC SOLS

Lyophilic Sols

1. Prepared by direct mixing with dispersion medium.
2. Little or no charge on particles.
3. Particles generally solvated.
4. Viscosity higher than dispersion medium; set to a gel.
5. Precipitated by high concentration of electrolytes.
6. Reversible.
7. Do not exhibit Tyndall effect.
8. Particles migrate to anode or cathode, or not at all.

Lyophobic Sols

1. Not prepared by direct mixing with the medium.
2. Particles carry positive or negative charge.
3. No solvation of particles.
4. Viscosity almost the same as of medium; do not set to a gel.
5. Precipitated by low concentration of electrolytes.
6. Irreversible.
7. Exhibit Tyndall effect.
8. Particles migrate to either anode or cathode.

EMULSIONS

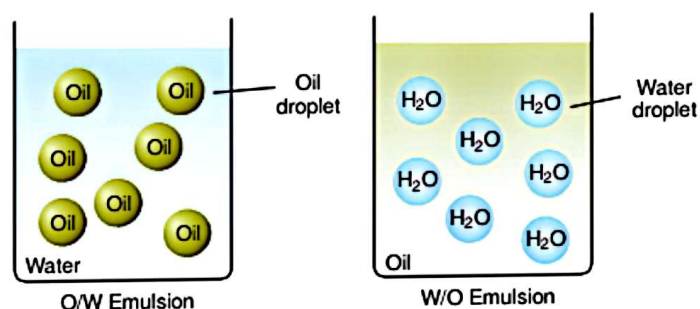
These are liquid-liquid colloidal systems. In other words, an emulsion may be defined as a **dispersion of finely divided liquid droplets in another liquid**.

Generally one of the two liquids is *water* and the other, which is immiscible with water, is designated as *oil*. Either liquid can constitute the dispersed phase.

Types of Emulsions

There are two types of emulsions.

(a) Oil-in-Water type (O/W type) ; (b) Water-in-Oil type (W/O type)



■ **Figure 22.27**
Two types of Emulsions.

Examples of Emulsions

- (1) Milk is an emulsion of O/W type. Tiny droplets of liquid fat are dispersed in water.
- (2) Stiff greases are emulsions of W/O type, water being dispersed in lubricating oil.

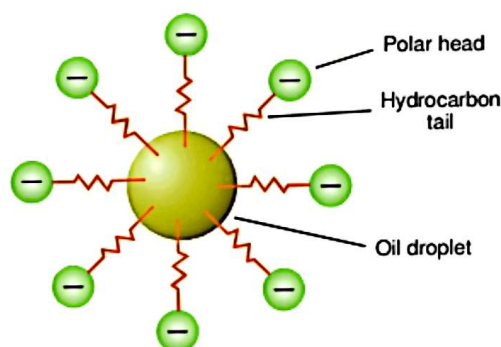
Preparation of Emulsions

The dispersal of a liquid in the form of an emulsion is called **emulsification**. This can be done by agitating a small proportion of one liquid with the bulk of the other. It is better accomplished by passing a mixture of the two liquid through a colloid mill known as **homogenizer**.

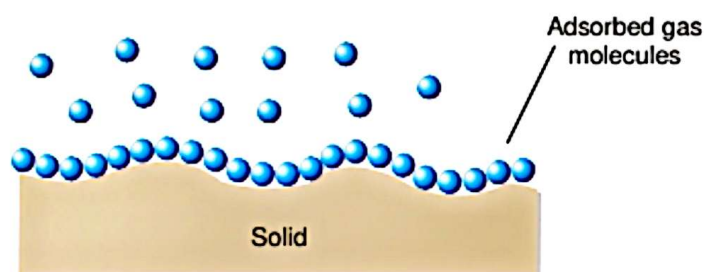
The emulsions obtained simply by shaking the two liquids are unstable. The droplets of the dispersed phase coalesce and form a separate layer. To have a stable emulsion, small amount of a third substance called the **Emulsifier** or **Emulsifying agent** is added during the preparation. This is usually a soap, synthetic detergent, or a hydrophilic colloid.

Role of Emulsifier

The emulsifier concentrates at the interface and reduces surface tension on the side of one liquid which rolls into droplets. Soap, for example, is made of a long hydrocarbon tail (oil soluble) with a polar head COO^-Na^+ (water soluble). In O/W type emulsion the tail is pegged into the oil droplet, while the head extends into water. Thus the soap acts as go-between and the emulsified droplets are not allowed to coalesce.



■ **Figure 22.28**
Role of emulsifier (Soap).



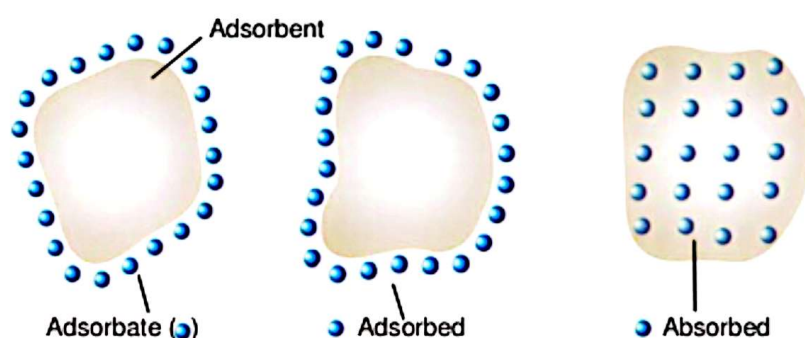
■ **Figure 23.1**
Adsorption of a gas at a solid surface.

The phenomenon of concentration of molecules of a gas or liquid at a solid surface is called adsorption.

The substance that deposits at the surface is called **Adsorbate** and the solid on whose surface the deposition occurs is called the **Adsorbent**.

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■ **Figure 23.2**
Adsorption versus absorption (illustration).

Examples of Adsorption

(1) **Adsorption of a dye by a charcoal.** If finely divided charcoal is stirred into a dilute solution of *methylene blue* (an organic dye), the depth of colour of the solution decreases appreciably. The dye molecules have been adsorbed by charcoal particles.

(2) **Adsorption of a gas by charcoal.** If a gas (SO_2 , Cl_2 , NH_3) is treated with powdered charcoal in a closed vessel, the gas pressure is found to decrease. The gas molecules concentrate on charcoal surface and are said to be adsorbed.

layer. However, above a certain pressure,

COMPARISON OF PHYSICAL ADSORPTION AND CHEMISORPTION

Physical adsorption

1. Caused by intermolecular van der Waal's forces.
2. Depends on nature of gas. Easily liquefiable gases are adsorbed readily.
3. Heat of adsorption is small (about 5 kcal mol⁻¹).
4. Reversible.
5. Occurs rapidly at low temperature; decreases with increasing temperature.
6. Increase of pressure increases adsorption; decrease of pressure causes desorption.
7. Forms multimolecular layers on adsorbent surface.

Chemisorption

1. Caused by chemical bond formation.
2. Much more specific than physical adsorption.
3. Heat of adsorption is large (20–100 kcal mol⁻¹).
4. Irreversible.
5. Increases with increase of temperature.
6. Change of pressure has no such effects.
7. Forms unimolecular layer.

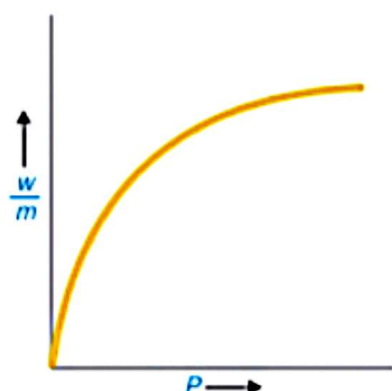
The relationship between the equilibrium pressure of a gas and its amount adsorbed on the solid adsorbent at any constant temperature is called an Adsorption isotherm. It may be given in the form of an equation or graphical curve.

Freundlich Adsorption Isotherm

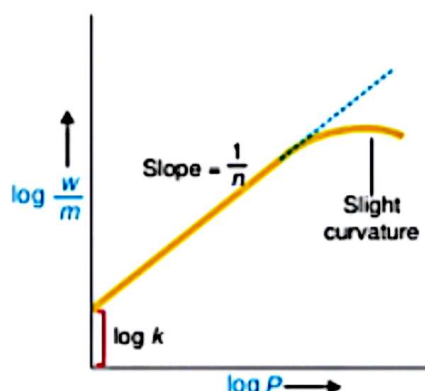
Freundlich proposed an empirical relation in the form of a mathematical equation.

$$\frac{w}{m} = k P^{1/n}$$

where w is the mass of the gas adsorbed on a mass m of adsorbent at a pressure P ; k and n are constants depending on the nature of the gas and the adsorbent and on temperature. This relation is generally represented in the form of a curve obtained by plotting the mass of the gas adsorbed per unit mass of adsorbent (w/m) against equilibrium pressure.



■ **Figure 23.6**
Freundlich Isotherm; a plot of mass of adsorbed gas per unit mass of adsorbent.



■ **Figure 23.7**
Plot of $\log w/m$ against $\log P$ shows slight curvature at higher pressures.

Freundlich Isotherm is not applicable at high pressures. Taking logarithms on both sides of Freundlich equation, we have

$$\log \frac{w}{m} = \log k + \frac{1}{n} \log P$$

This is equation for a straight line. Thus a plot of $\log (w/m)$ against $\log P$ should be a straight line with slope $1/n$ and intercept $\log k$. However, it is actually found that the plots were straight lines at low pressures, while at higher pressure they showed a slight curvature, especially at low temperature. This indicated that Freundlich equation is approximate and does not apply to adsorption of gases by solids at higher pressures.

LANGMUIR ADSORPTION ISOTHERM

Langmuir (1916) derived a simple adsorption isotherm based on theoretical considerations. It was named after him.

Assumptions

Langmuir made the following assumptions.

- (1) The layer of the gas adsorbed on the solid adsorbent is one-molecule thick.
- (2) The adsorbed layer is uniform all over the adsorbent.
- (3) There is no interaction between the adjacent adsorbed molecules.

Derivation of Langmuir Isotherm

Langmuir considered that the gas molecules strike a solid surface and are thus adsorbed. Some

Theory of Dilute Solutions



In this chapter we will restrict our discussion to solutions in which a solid is dissolved in a liquid. The solid is referred to as the **solute** and the liquid as the **solvent**.

COLLIGATIVE PROPERTIES

Dilute solutions containing non-volatile solute exhibit the following properties :

- (1) Lowering of the Vapour Pressure
- (2) Elevation of the Boiling Point
- (3) Depression of the Freezing Point
- (4) Osmotic Pressure

The essential feature of these properties is that they depend *only* on the number of solute particles present in solution. Being closely related to each other through a common explanation, these have been grouped together under the class name **Colligative Properties** (Greek *colligatus* = Collected together).

A colligative property may be defined as one which depends on the number of particles in solution and not in any way on the size or chemical nature of the particles.

Consequent to the above definition, each colligative property is exactly related to any other. Thus if one property is

substances.

LOWERING OF VAPOUR PRESSURE : RAOULT'S LAW

The vapour pressure of a pure solvent is decreased when a non-volatile solute is dissolved in it. If p is the vapour pressure of the solvent and p_s that of the solution, the lowering of vapour pressure is $(p - p_s)$. This lowering of vapour pressure relative to the vapour pressure of the pure solvent is termed the **Relative lowering of Vapour pressure**. Thus,

$$\text{Relative Lowering of Vapour Pressure} = \frac{p - p_s}{p}$$

As a result of extensive experimentation, Raoult (1886) gave an empirical relation connecting the relative lowering of vapour pressure and the concentration of the solute in solution. This is now referred to as the **Raoult's Law**. It states that : **the relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.**

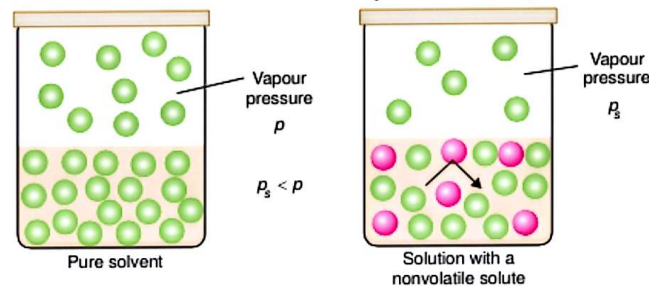
Raoult's Law can be expressed mathematically in the form :

$$\frac{p - p_s}{p} = \frac{n}{n + N}$$

where n = number of moles or molecules of solute
 N = number of moles or molecules of solvent.

Derivation of Raoult's Law

The vapour pressure of the pure solvent is caused by the number of molecules evaporating from its surface. When a nonvolatile solute is dissolved in solution, the presence of solute molecules in the surface blocks a fraction of the surface where no evaporation can take place.



■ Figure 15.1

Lowering of vapour pressure by a nonvolatile solute. The particles of the solute block the escape of solvent molecules from the surface of the solution.

This causes the lowering of the vapour pressure. The vapour pressure of the solution is, therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction. That is,

$$p_s \propto \frac{N}{n + N}$$

where N = moles of solvent and n = moles of solute.

$$\text{or} \quad p_s = k \frac{N}{n + N} \quad \dots(1)$$

k being proportionality factor.

In case of pure solvent $n = 0$ and hence

$$\text{Mole fraction of solvent} = \frac{N}{n + N} = \frac{N}{0 + N} = 1$$

Now from equation (1), the vapour pressure $p = k$

Therefore the equation (1) assumes the form

$$p_s = p \frac{N}{n + N}$$

or

$$\frac{p_s}{p} = \frac{N}{n + N}$$

$$1 - \frac{p_s}{p} = 1 - \frac{N}{n + N}$$

$$\frac{p - p_s}{p} = \frac{n}{n + N}$$

This is Raoult's Law.

SOLVED PROBLEM Calculate the vapour pressure lowering caused by the addition of 100 g of

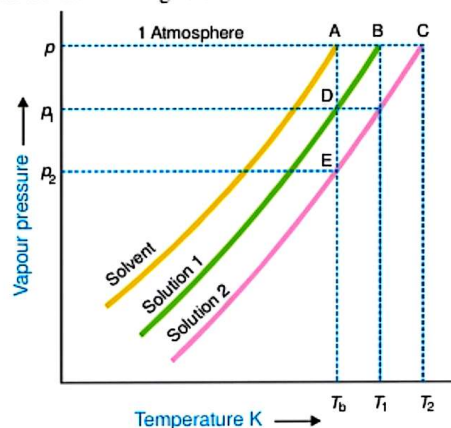
ELEVATION OF BOILING POINT

Relation between Elevation of Boiling Point and Lowering of Vapour-pressure

When a liquid is heated, its vapour pressure rises and when it equals the atmospheric pressure, the liquid boils. The addition of a non volatile solute lowers the vapour pressure and consequently elevates the boiling point as the solution has to be heated to a higher temperature to make its vapour pressure become equal to atmospheric pressure. If T_b is the boiling point of the solvent and T is the boiling point of the solution, the difference in the boiling points (ΔT) is called the **elevation of boiling point**.

$$T - T_b = \Delta T$$

The vapour pressure curves of the pure solvent, and solutions (1) and (2) with different concentrations of solute are shown in Fig.15.5.



■ Figure 15.5

Ostwald-Walker method of measuring the relative lowering of vapour pressure.

For dilute solutions, the curves BD and CE are parallel and straight lines approximately. Therefore for similar triangles ACE and ABD , we have

$$\frac{AB}{AC} = \frac{AD}{AE}$$

$$\frac{T_1 - T_b}{T_2 - T_b} = \frac{p - p_1}{p - p_2}$$

or

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where $p - p_1$ and $p - p_2$ are lowering of vapour pressure for solution 1 and solution 2 respectively. Hence the elevation of boiling point is directly proportional to the lowering of vapour pressure.

$$\text{or} \quad \Delta T \propto p - p_s \quad \dots(1)$$

Determination of Molecular Mass from Elevation of Boiling Point

Since p is constant for the same solvent at a fixed temperature, from (1) we can write

$$\Delta T \propto \frac{p - p_s}{p} \quad \dots(2)$$

But from Raoult's Law for dilute solutions,

$$\frac{p - p_s}{p} \propto \frac{wM}{Wm} \quad \dots(3)$$

Since M (mol mass of solvent) is constant, from (3)

$$\frac{p - p_s}{p} \propto \frac{w}{Wm} \quad \dots(4)$$

From (2) and (4)

$$\Delta T \propto \frac{w}{m} \times \frac{1}{W}$$

$$\text{or} \quad \Delta T = K_b \times \frac{w}{m} \times \frac{1}{W} \quad \dots(5)$$

where K_b is a constant called **Boiling point constant** or **Ebulioscopic constant of molal elevation constant**. If $w/m = 1$, $W = 1$, $K_b = \Delta T$. Thus,

Molal elevation constant may be defined as the boiling-point elevation produced when 1 mole of solute is dissolved in one kg (1000 g) of the solvent.

If the mass of the solvent (W) is given in grams, it has to be converted into kilograms. Thus the

Ideal Solutions and Deviations from Raoult's Law

A solution which obeys Raoult's law strictly is called an **Ideal solution**. A solution which shows deviations from Raoult's law is called a **Nonideal or Real solution**.

Suppose the molecules of the solvent and solute are represented by A and B respectively. Now let γ_{AB} be the attractive force between A and B, and γ_{AA} between A and A. If

$$\gamma_{AB} = \gamma_{AA}$$

the solution will show the same vapour pressure as predicted by Raoult's law and it is an ideal

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solution. However, if

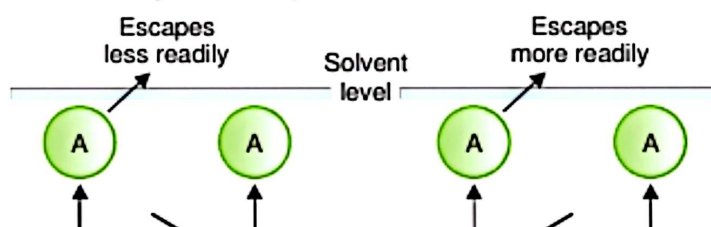
$$\gamma_{AB} > \gamma_{AA}$$

molecule A will escape less readily and the vapour pressure will be less than that predicted by Raoult's law (**Negative deviation**). On the other hand, if

$$\gamma_{AB} < \gamma_{AA}$$

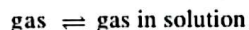
A molecule will escape from the solution surface more readily and the vapour pressure of the solution will be higher than predicted by Raoult's law (**Positive deviation**).

In very dilute solutions of nonelectrolytes, the solvent and solute molecules are very much alike in both molecular size and molecular attractions. Thus such solutions tend to approach the ideal behaviour and obey Raoult's law fairly accurately.



HENRY'S LAW

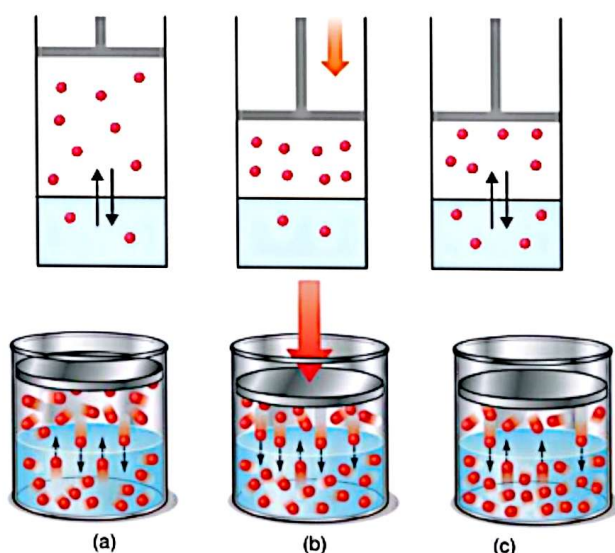
The solubility of a gas in a solvent depends on the pressure and the temperature. When a gas is enclosed over its saturated solution, the following equilibrium exists.



If pressure is increased on the system, the equilibrium will move in the direction which will reduce the pressure (Le Chatelier Principle). The pressure can be reduced by more gas dissolving in solvent. Thus solubility or concentration of a gas in a given solvent is increased with increase of pressure. A kinetic molecular explanation of the effect of pressure on gas-solution system is illustrated in Fig. 14.2.

The relationship between pressure and solubility of a gas in a particular solvent was investigated by William Henry (1803). He gave a generalisation which is known as **Henry's Law**. It may be stated as : **for a gas in contact with a solvent at constant temperature, concentration of the gas that dissolves in the solvent is directly proportional to the pressure of the gas.**

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■ Figure 14.2

(a) Equilibrium exists between the gas molecules and those in solution;
(b) When pressure is applied, gas volume decreases and concentration in gas phase increases; (c) More gas molecules pass into solution to reestablish equilibrium whereby the pressure above the gas is decreased.

Mathematically, Henry's Law may be expressed as

$$C \propto P$$

or

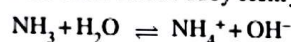
$$C = k P$$

where P = pressure of the gas; C = concentration of the gas in solution; and k = proportionality constant known as **Henry's Law Constant**. The value of k depends on the nature of the gas and solvent, and the units of P and C used.

Limitations of Henry's Law

It applies closely to gases with nearly ideal behaviour.

- (1) at moderate temperature and pressure.
- (2) if the solubility of the gas in the solvent is low.
- (3) the gas does not react with the solvent to form a new species. Thus ammonia (or HCl) which react with water do not obey Henry's Law.



- (4) the gas does not associate or dissociate on dissolving in the solvent.

$$\Delta G^\circ = -RT \ln K \quad \dots (iv)$$

or
$$\Delta G^\circ = -2.303 RT \log K \quad \dots (v)$$

The equation (iv) is also called van't Hoff Isotherm. It may be used to calculate the change in free energy of a reaction in the standard (ΔG°) from the equilibrium constant and vice-versa.

The sign of ΔG° indicates whether the forward or reverse reaction is spontaneous. Considering the equation (iv), we can have three possibilities depending on the sign of ΔG° for the reaction.

- (1) If **ΔG° is negative**, $\log K$ must be positive and the reaction proceeds spontaneously in the forward reaction.
- (2) If **ΔG° is positive**, $\log K$ must be negative and K is less than one. The reverse reaction is then spontaneous.
- (3) If **$\Delta G^\circ = 0$** , $\log K = 0$ and $K = 1$. The reaction is at equilibrium.

Measuring the Magnitude of the Free Energy Change

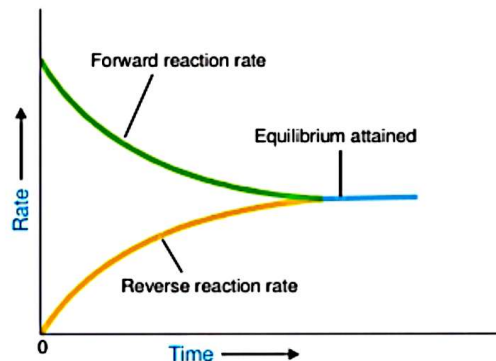
NATURE OF CHEMICAL EQUILIBRIUM : ITS DEFINITION

Let us consider the reaction



If we start with A and B in a closed vessel, the forward reaction proceeds to form C and D. The concentrations of A and B decrease and those of C and D increase continuously. As a result the rate of forward reaction also decreases and the rate of the reverse reaction increases. Eventually, the rate of the two opposing reactions equals and the system attains a *state of equilibrium*. Thus **Chemical equilibrium** may be defined as: **the state of a reversible reaction when the two opposing reactions occur at the same rate and the concentrations of reactants and products do not change with time.**

Furthermore, the true equilibrium of a reaction can be attained from both sides. Thus the equilibrium concentrations of the reactants and products are the same whether we start with A and B, or C and D.

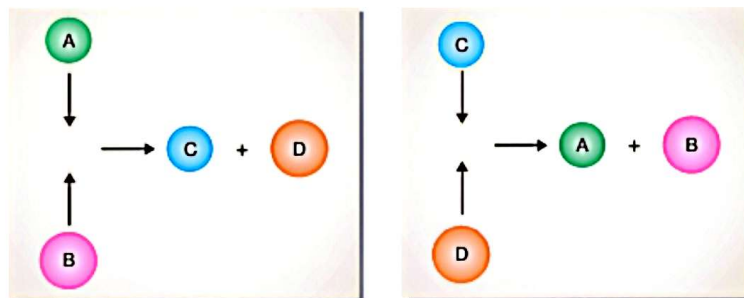


■ Figure 17.1

At equilibrium the forward reaction rate equals the reverse reaction rate.

Chemical Equilibrium is Dynamic Equilibrium

We have shown above that as the reaction $A + B \rightleftharpoons C + D$ attains equilibrium, the concentrations of A and B, as also of C and D remain constant with time. Apparently it appears that the equilibrium is dead. But it is not so. The equilibrium is dynamic. Actually, the forward and the reverse reactions are taking place at equilibrium but the concentrations remain unchanged.



■ Figure 17.2

Molecules of A and B colliding to give C and D, and those of C and D colliding to give A and B.

The dynamic nature of chemical equilibrium can be easily understood on the basis of the kinetic molecular model. The molecules of A and B in the equilibrium mixture collide with each other to form C and D. Likewise C and D collide to give back A and B. The collisions of molecules in a closed system is a ceaseless phenomenon. Therefore collisions of A and B giving C and D (Forward reaction) and collisions of C and D giving back A and B (reverse reaction) continue to occur even at equilibrium, while concentrations remain unchanged.

CHARACTERISTICS OF CHEMICAL EQUILIBRIUM

Before we take up the mathematical study of chemical equilibria, let us understand the chemical characteristics of chemical equilibrium.

(1) Constancy of concentrations

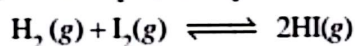
When a chemical equilibrium is established in a closed vessel at constant temperature, concentrations of the various species in the reaction mixture become constant.

The reaction mixture at equilibrium is called **Equilibrium mixture**.

The concentrations at equilibrium are called **Equilibrium concentrations**. The equilibrium concentrations are represented by square brackets with subscript *eq* or *eqm*, $[]_{eq}$. Thus $[A]_{eq}$ denotes the equilibrium concentration of substance A in moles per litre. In modern practice the subscript *eq* is not used.

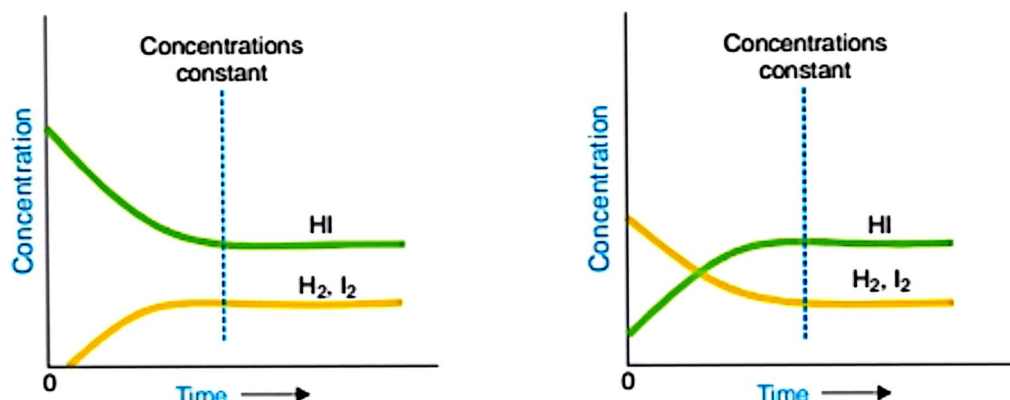
(2) Equilibrium can be initiated from either side

The state of equilibrium of a reversible reaction can be approached whether we start with reactants or products, for example, the equilibrium



is established if we start the reaction with H_2 and I_2 , or 2 HI.

Fig. 17.3. Shows whether we start with 1 mole of I_2 and 1 mole of H_2 , or with 2 moles of HI in a vessel of the same volume, the same mixture of H_2 , I_2 and HI is obtained.



■ Figure 17.3

The left graph shows how equilibrium is attained for start with H_2 and I_2 .

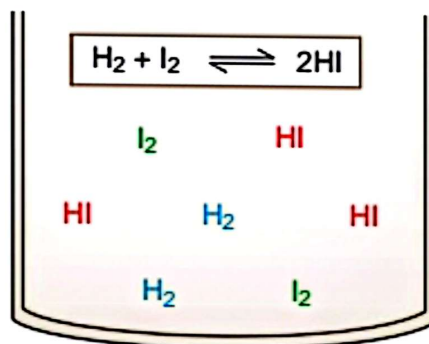
The right graph depicts the attainment of equilibrium for start with 2HI.

Equilibrium concentrations in both cases are the same.

(3) Equilibrium cannot be Attained in an Open Vessel

The equilibrium can be established only if the reaction vessel is closed and no part of the reactants or products is allowed to escape out. In an open vessel, the gaseous reactants and/or products may escape into the atmosphere leaving behind no possibility of attaining equilibrium.

However, the equilibrium can be attained when all the reactants and products are in the same phase *i.e.*, ethanol and ethanoic acid.



■ **Figure 17.4**
A chemical equilibrium between H_2 , I_2 and HI .

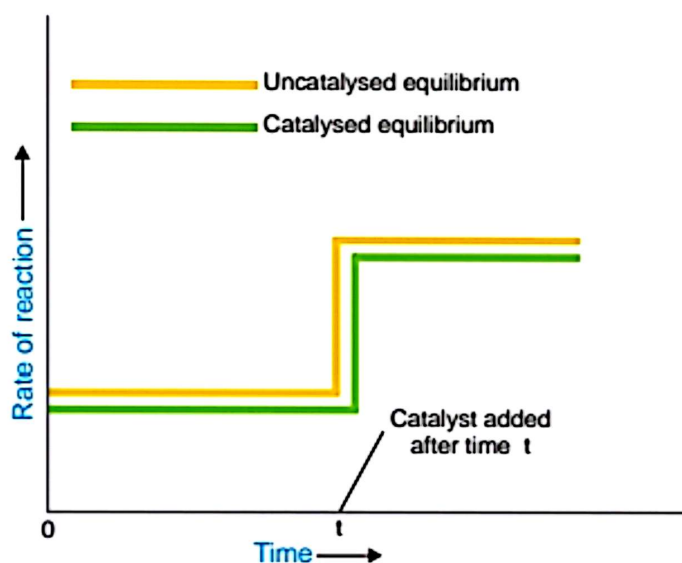
(4) A catalyst cannot change the equilibrium point

When a catalyst is added to a system in equilibrium, it speeds up the rate of both the forward and the reverse reaction to an equal extent. Therefore a catalyst cannot change the equilibrium point except that it is achieved earlier. This enhances the rate of the reaction.

The rapid increase of the rate of an equilibrium reaction on the addition of a catalyst is illustrated in Fig. 17.5.

(5) Value of Equilibrium Constant does not depend upon the initial concentration of reactants

It has been found that equilibrium constant must be the same when the concentrations of reacting species are varied over a wide range.



■ **Figure 17.5**
The vertical part of the curve indicates rapid increase of rate of reaction on addition of catalyst.

proceeds without change in free energy *i.e.* $\Delta G = 0$.

LAW OF MASS ACTION

Two Norwegian chemists, Guldberg and Waage, studied experimentally a large number of equilibrium reactions. In 1864, they postulated a generalisation called the **Law of Mass action**. It states that : **the rate of a chemical reaction is proportional to the active masses of the reactants.**

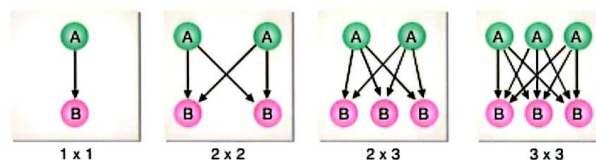
By the term 'active mass' is meant the molar concentration *i.e.*, number of moles per litre. It is expressed by enclosing the formula of the substance in square brackets. Thus a gas mixture containing 1 g of hydrogen (H_2) and 127 g of iodine (I_2) per litre has the concentrations

$$[H_2] = 0.5 \text{ and } [I_2] = 0.5$$

Explanation of the Law of Mass Action Based on the Molecular Collision Theory

We assume that a chemical reaction occurs as a result of the collisions between the reacting molecules. Although some of these collisions are ineffective, the chemical change produced is proportional to the number of collisions actually taking place. Thus at a fixed temperature the rate of a reaction is determined by the number of collisions between the reactant molecules present in unit volume.

Let us consider four boxes of one cubic centimetre volume; containing different number of reacting molecules A and B (Fig 17.6). They undergo collisions to form the products C and D, the rate of reaction being governed by the number of possible collisions between them.



■ **Figure 17.6**
Possible collisions between molecules of the reactants A and B.

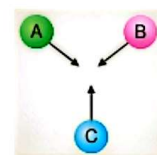
In the first box there is one molecule of A and one molecule of B, and the possibilities of collision at any instant is $1 \times 1 = 1$. In the second box, the number of molecules of A and B are two each and the possibilities of collisions are $2 \times 2 = 4$. In the third box, there are two molecules of A and three molecules of B, and the number of possibilities of collisions between A and B are increased to $3 \times 2 = 6$. In the fourth box, we have three molecules of each of A and B and the total number of collisions between them is $3 \times 3 = 9$. Thus in general, we can say that the possibilities of collisions between the reacting molecules A and B are equal to the product of the number of molecules of each species per unit volume. Since the rate of reaction is determined by molecular impacts, it is proportional to moles per unit volume *i.e.*, molar concentration. Thus we can write

$$\begin{aligned} \text{Rate of reaction} &\propto [A][B] \\ &= k [A][B] \end{aligned}$$

In a reaction, $A + B + C \longrightarrow$ the number of collisions between A, B and C depends on the concentration of each reactant (Fig. 17.7). Therefore,

$$\text{Rate of reaction} \propto [A][B][C] = k [A][B][C]$$

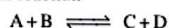
From the above considerations, it stands proved that the rate of a reaction is proportional to the molar concentrations of the reactants.



■ **Figure 17.7**

EQUILIBRIUM CONSTANT : EQUILIBRIUM LAW

Let us consider a general reaction



and let $[A]$, $[B]$, $[C]$ and $[D]$ represent the molar concentrations of A, B, C and D at the equilibrium point. According to the Law of Mass action.

$$\text{Rate of forward reaction} \propto [A][B] = k_1 [A][B]$$

$$\text{Rate of reverse reaction} \propto [C][D] = k_2 [C][D]$$

where k_1 and k_2 are rate constants for the forward and reverse reactions.

At equilibrium, rate of forward reaction = rate of reverse reaction.

Therefore,

$$k_1 [A][B] = k_2 [C][D]$$

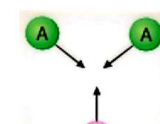
$$\text{or} \quad \frac{k_1}{k_2} = \frac{[C][D]}{[A][B]} \quad \dots (1)$$

At any specific temperature k_1/k_2 is constant since both k_1 and k_2 are constants. The ratio k_1/k_2 is called **Equilibrium constant** and is represented by the symbol K_c , or simply k . The subscript 'c' indicates that the value is in terms of concentrations of reactants and products. The equation (1) may be written as

$$\text{Equilibrium constant } K_c = \frac{[C][D]}{[A][B]}$$

Product concentrations
Reactant concentrations

This equation is known as the **Equilibrium constant expression or Equilibrium law**.





EQUILIBRIUM CONSTANT EXPRESSION IN TERMS OF PARTIAL PRESSURES

When all the reactants and products are gases, we can also formulate the equilibrium constant expression in terms of partial pressure. The relationship between the partial pressure (p) of any one gas in the equilibrium mixture and the molar concentration follows from the general ideal gas equation

$$pV = nRT \quad \text{or} \quad p = \left(\frac{n}{V} \right) RT$$

The quantity $\frac{n}{V}$ is the number of moles of the gas per unit volume and is simply the molar concentration. Thus,

$$p = (\text{Molar concentration}) \times RT$$

i.e., the partial pressure of a gas in the equilibrium mixture is directly proportional to its molar concentration at a given temperature. Therefore, we can write the equilibrium constant expression in terms of partial pressure instead of molar concentrations. For a general reaction



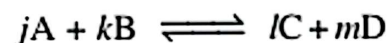
the equilibrium law or the equilibrium constant may be written as

$$K_p = \frac{(p_Y)^y (p_Z)^z}{(p_L)^l (p_M)^m}$$

Here K_p is the equilibrium constant, the subscript p referring to partial pressure. Partial pressures are expressed in atmospheres.

HOW K_c AND K_p ARE RELATED?

Let us consider a general reaction



where all reactants and products are gases. We can write the equilibrium constant expression in terms of partial pressures as

$$K_p = \frac{(p_C)^l (p_D)^m}{(p_A)^j (p_B)^k} \quad \dots(1)$$

Assuming that all these gases constituting the equilibrium mixture obey the ideal gas equation, the partial pressure (p) of a gas is

$$p = \left(\frac{n}{V} \right) RT$$

Where $\frac{n}{V}$ is the molar concentration. Thus the partial pressures of individual gases, A, B, C and D are:

$$p_A = [A] RT; p_B = [B] RT; p_C = [C] RT; p_D = [D] RT$$

Substituting these values in equation (1), we have

$$K_p = \frac{[C]^l (RT)^l [D]^m (RT)^m}{[A]^j (RT)^j [B]^k (RT)^k}$$

or

$$K_p = \frac{[C]^l [D]^m}{[A]^j [B]^k} \times \frac{(RT)^{l+m}}{(RT)^{j+k}}$$

$$K_p = K_c \times (RT)^{(l+m)-(j+k)}$$

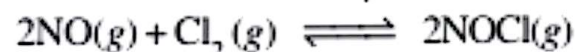
$$\therefore K_p = K_c \times (RT)^{\Delta n} \quad \dots(2)$$

where $\Delta n = (l + m) - (j + k)$, the difference in the sums of the coefficients for the gaseous products and reactants.

From the expression (2) it is clear that when $\Delta n = 0$, $K_p = K_c$.

$$= 1.9 \times 10^3$$

SOLVED PROBLEM 2. The value of K_p at 25°C for the reaction



is $1.9 \times 10^3 \text{ atm}^{-1}$. Calculate the value of K_c at the same temperature.

SOLUTION

We can write the general expression as

$$K_p = K_c (RT)^{\Delta n} \text{ or } K_c = \frac{K_p}{(RT)^{\Delta n}}$$

Here,

$$T = 25 + 273 = 298 \text{ K}$$

$$R = 0.0821$$

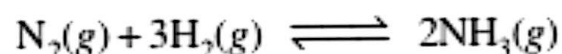
$$\Delta n = 2 - (2 + 1) = -1$$

$$K_p = 1.9 \times 10^3$$

Substituting these values in the general expression

$$\begin{aligned} K_c &= \frac{1.9 \times 10^3}{(0.0821 \times 298)^{-1}} \\ &= 4.6 \times 10^4 \end{aligned}$$

SOLVED PROBLEM 2. Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2 , 2.10 moles of H_2 and 0.298 mole of NH_3 are found to be present. Find the value of K_c for the reaction



at 500°C.

SOLUTION

The equilibrium concentrations are obtained by dividing the number of moles of each reactant and product by the volume, 5.00 litres. Thus,

$$[N_2] = 3.00 \text{ mole}/5.00 \text{ L} = 0.600 \text{ M}$$

$$[H_2] = 2.10 \text{ mole}/5.00 \text{ L} = 0.420 \text{ M}$$

$$[NH_3] = 0.298 \text{ mole}/5.00 \text{ L} = 0.0596 \text{ M}$$

Substituting these concentrations (not number of moles) in the equilibrium constant expression, we get the value of K_c .

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.600)(0.420)^3} = 0.080$$

Thus, for the reaction of H_2 and N_2 to form NH_3 at 500°C, we can write

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \mathbf{0.080}$$

LE CHATELIER'S PRINCIPLE

In 1884, the French Chemist Henry Le Chatelier proposed a general principle which applies to all systems in equilibrium. This important principle called the **Le Chatelier's principle** may be stated as : **when a stress is applied on a system in equilibrium, the system tends to adjust itself so as to reduce the stress.**

There are three ways in which the stress can be caused on a chemical equilibrium:

- (1) Changing the concentration of a reactant or product.
- (2) Changing the pressure (or volume) of the system.
- (3) Changing the temperature.

Thus when applied to a chemical reaction in equilibrium, Le Chatelier's principle can be stated as: **if a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimise the change.**

Now we proceed to illustrate the above statement by taking examples of each type of stress.

EFFECT OF A CHANGE IN CONCENTRATION

We can restate Le Chatelier's principle for the special case of concentration changes: **when concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.**

Addition of an inert gas

Let us add an *inert gas* to an equilibrium mixture while the volume of the reaction vessel remains the same. The addition of the inert gas increases the total pressure but the partial pressures of the reactants and products are not changed. Thus **the addition of an inert gas has no effect on the position of the equilibrium.**

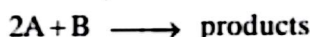
RATE LAWS

At a fixed temperature the rate of a given reaction depends on concentration of reactants. The exact relation between concentration and rate is determined by measuring the reaction rate with different initial reactant concentrations. By a study of numerous reactions it is shown that : **the rate of a reaction is directly proportional to the reactant concentrations, each concentration being raised to some power.**

Thus for a substance A undergoing reaction,

$$\begin{array}{ll} \text{rate} & \text{rate} \propto [A]^n \\ \text{or} & \text{rate} = k [A]^n \end{array} \quad \dots(1)$$

For a reaction



the reaction rate with respect to A or B is determined by varying the concentration of one reactant, keeping that of the other constant. Thus the rate of reaction may be expressed as

$$\text{rate} = k [A]^m [B]^n \quad \dots(2)$$

Expressions such as (1) and (2) tell the relation between the rate of a reaction and reactant concentrations.

An expression which shows how the reaction rate is related to concentrations is called the rate law or rate equation.

The power (exponent) of concentration n or m in the rate law is usually a small whole number integer (1, 2, 3) or fractional. The proportionality constant k is called the **rate constant** for the reaction.

Examples of rate law :

	REACTIONS	RATE LAW
(1)	$2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$	$\text{rate} = k [\text{N}_2\text{O}_5]$
(2)	$\text{H}_2 + \text{I}_2 \longrightarrow 2\text{HI}$	$\text{rate} = k [\text{H}_2] [\text{I}_2]$
(3)	$2\text{NO}_2 \longrightarrow 2\text{NO} + \text{O}_2$	$\text{rate} = k [\text{NO}_2]^2$
(4)	$2\text{NO} + 2\text{H}_2 \longrightarrow \text{N}_2 + 2\text{H}_2\text{O}$	$\text{rate} = k [\text{H}_2] [\text{NO}]^2$

In these rate laws where the quotient or concentration is not shown, it is understood to be 1. That is $[\text{H}_2]^1 = [\text{H}_2]$.

It is apparent that the rate law for a reaction must be determined by experiment. It cannot be written by merely looking at the equation with a background of our knowledge of Law of Mass Action. However, for some elementary reactions the powers in the rate law may correspond to coefficients in the chemical equation. **But usually the powers of concentration in the rate law are different from coefficients.** Thus for the reaction (4) above, the rate is found to be proportional to $[\text{H}_2]$ although the quotient of H_2 in the equation is 2. For NO the rate is proportional to $[\text{NO}]^2$ and power '2' corresponds to the coefficient.

ORDER OF A REACTION

The order of a reaction is defined as the sum of the powers of concentrations in the rate law.

Let us consider the example of a reaction which has the rate law

$$\text{rate} = k [A]^m [B]^n \quad \dots(1)$$

The order of such a reaction is $(m + n)$.

Examples of reaction order :

RATE LAW	REACTION ORDER
$\text{rate} = k [\text{N}_2\text{O}_5]$	1
$\text{rate} = k [\text{H}_2] [\text{I}_2]$	$1 + 1 = 2$
$\text{rate} = k [\text{NO}_2]^2$	2
$\text{rate} = k [[\text{H}_2] [\text{NO}]^2]$	$1 + 2 = 3$
$\text{rate} = k [\text{CHCl}_3] [\text{Cl}_2]^{1/2}$	$1 + \frac{1}{2} = 1\frac{1}{2}$

Reactions may be classified according to the order. If in the rate law (1) above

$m + n = 1$, it is **first order reaction**

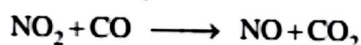
$m + n = 2$, it is **second order reaction**

$m + n = 3$, it is **third order reaction**

ZERO ORDER REACTION

A reactant whose concentration does not affect the reaction rate is not included in the rate law. In effect, the concentration of such a reactant has the power 0. Thus $[\text{A}]^0 = 1$.

A zero order reaction is one whose rate is independent of concentration. For example, the rate law for the reaction



at 200°C is

$$\text{rate} = k [\text{NO}_2]^2$$

Here the rate does not depend on $[\text{CO}]$, so this is not included in the rate law and the power of $[\text{CO}]$ is understood to be zero. The reaction is **zeroth order** with respect to CO . The reaction is second order with respect to $[\text{NO}_2]$. The overall reaction order is $2 + 0 = 2$.

MOLECULARITY OF A REACTION

Chemical reactions may be classed into two types :

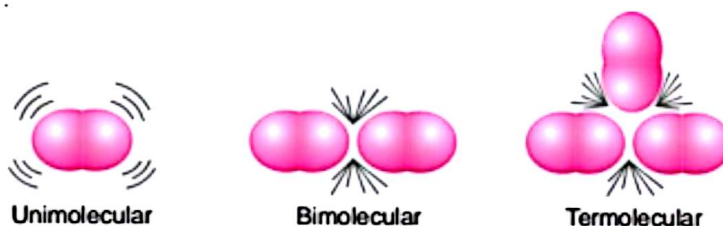
- Elementary reactions
- Complex reactions

An **elementary reaction** is a simple reaction which occurs in a single step.

A **complex reaction** is that which occurs in two or more steps.

Molecularity of an Elementary Reaction

The molecularity of an elementary reaction is defined as : the number of reactant molecules involved in a reaction.



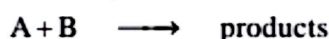
Differences Between Order and Molecularity

Order of a Reaction	Molecularity of a Reaction
1. It is the sum of powers of the concentration terms in the rate law expression. 2. It is an experimentally determined value. 3. It can have fractional value. 4. It can assume zero value. 5. Order of a reaction can change with the conditions such as pressure, temperature, concentration.	1. It is number of reacting species undergoing simultaneous collision in the elementary or simple reaction. 2. It is a theoretical concept. 3. It is always a whole number. 4. It can not have zero value. 5. Molecularity is invariant for a chemical equation.

PSEUDO-ORDER REACTIONS

A reaction in which one of the reactants is present in a large excess shows an order different from the actual order. **The experimental order which is not the actual one is referred to as the pseudo order.** Since for elementary reactions molecularity and order are identical, pseudo-order reactions may also be called **pseudo molecular reactions**.

Let us consider a reaction



in which the reactant B is present in a large excess. Since it is an elementary reaction, its rate law can be written as

$$\text{rate} = k [A] [B]$$

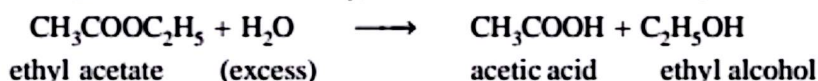
As B is present in large excess, its concentration remains practically constant in the course of reaction. Thus the rate law can be written as

$$\text{rate} = k' [A]$$

where the new rate constant $k' = k [B]$. Thus the actual order of the reaction is second-order but in practice it will be first-order. Therefore, the reaction is said to have a **pseudo-first order**.

Examples of Pseudo-order Reactions

- (1) **Hydrolysis of an ester.** For example, ethyl acetate upon hydrolysis in aqueous solution using a mineral acid as catalyst forms acetic acid and ethyl alcohol.

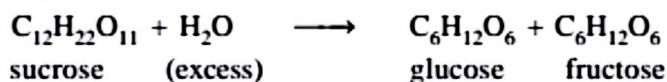


Here a large excess of water is used and the rate law can be written as

$$\begin{aligned} \text{rate} &= k [\text{CH}_3\text{COOH}] [\text{H}_2\text{O}] \\ &= k' [\text{CH}_3\text{COOH}] \end{aligned}$$

The reaction is actually second-order but in practice it is found to be first-order. Thus it is a pseudo-first order reaction.

- (2) **Hydrolysis of sucrose.** Sucrose upon hydrolysis in the presence of a dilute mineral acid gives glucose and fructose.



If a large excess of water is present, $[\text{H}_2\text{O}]$ is practically constant and the rate law may be written as

FIRST ORDER REACTIONS

Let us consider a first order reaction



Suppose that at the beginning of the reaction ($t = 0$), the concentration of A is a moles litre⁻¹. If after time t , x moles of A have changed, the concentration of A is $a - x$. We know that for a first order reaction, the rate of reaction, dx/dt , is directly proportional to the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k(a - x)$$

$$\text{or} \quad \frac{dx}{a - x} = k dt \quad \dots(1)$$

Integration of the expression (1) gives

$$\int \frac{dx}{a - x} = \int k dt$$

$$\text{or} \quad -\ln(a - x) = kt + I \quad \dots(2)$$

where I is the constant of integration. The constant k may be evaluated by putting $t = 0$ and $x = 0$.

Thus,

$$I = -\ln a$$

Substituting for I in equation (2)

$$\ln \frac{a}{a - x} = kt \quad \dots(3)$$

$$\text{or} \quad k = \frac{1}{t} \ln \frac{a}{a - x}$$

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Changing into common logarithms

$$k = \frac{2.303}{t} \log \frac{a}{a - x} \quad \dots(4)$$

The value of k can be found by substituting the values of a and $(a - x)$ determined experimentally at time interval t during the course of the reaction.

Sometimes the integrated rate law in the following form is also used :

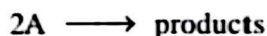
$$k = \frac{2.303}{t_2 - t_1} \log \frac{(a - x_1)}{(a - x_2)}$$

where x_1 and x_2 are the amounts decomposed at time intervals t_1 and t_2 respectively from the start.

Examples of First order Reactions

SECOND ORDER REACTIONS

Let us take a second order reaction of the type



Suppose the initial concentration of A is a moles litre⁻¹. If after time t , x moles of A have reacted, the concentration of A is $(a - x)$. We know that for such a second order reaction, rate of reaction is proportional to the square of the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k(a - x)^2 \quad \dots(1)$$

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where k is the rate constant, Rearranging equation (1), we have

$$\frac{dx}{(a - x)^2} = k dt \quad \dots(2)$$

On integration, it gives

$$\frac{1}{a - x} = kt + I \quad \dots(3)$$

where I is integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = \frac{1}{a} \quad \dots(4)$$

Substituting for I in equation (3)

$$\begin{aligned} \frac{1}{a - x} &= kt + \frac{1}{a} \\ kt &= \frac{1}{a - x} - \frac{1}{a} \\ k &= \frac{1}{t} \cdot \frac{x}{a(a - x)} \end{aligned}$$

Thus

This is the integrated rate equation for a second order reaction.

Examples of Second order Reaction

Hydrolysis of an Ester by NaOH. This is typical second order reaction.

